

Enhancing Skin Cancer Detection

A Fine-Tuned ResNet50 Transfer Learning Framework with Decision Threshold Calibration

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Motivation & Problem Statement

- Malignant and benign skin lesions can look very similar, but they need different medical treatment.
- Skin lesion images can have many variations, such as differences in color, texture, shape, border, and lighting. Things like hair, shadows, and other image artifacts can also appear in the image. These variations make automatic classification more challenging for deep learning models.
- Previous research used a hybrid approach to solve this problem. For example, the reference work used DarkNet53 to extract deep features, then WPCA for feature reduction, followed by Random Forest for classification. This approach achieved 88.01% accuracy, providing a useful benchmark for comparison.
- Our objective is to develop a simpler end-to-end approach. Instead of extracting features and then using a separate classical machine-learning classifier, we fine-tune a CNN directly for the classification task. The main goal is to determine whether this directly fine-tuned CNN can achieve higher accuracy than the 88.01% benchmark while keeping the overall pipeline simpler.

From Handcrafted Features to Deep Transfer Learning

Classical Approaches

- Used handcrafted lesion features such as border, texture, and asymmetry.
- These features were then classified using SVM, KNN, or Random Forest.
- Main limitation: performance depends heavily on the quality of manually designed features.

Deep Feature + Hybrid Classifiers

- Recent studies use CNNs such as ShuffleNet, SqueezeNet, DarkNet53, and Xception to extract deep features.
- These features are then reduced using WPCA and classified using Random Forest or SVM.
- DarkNet53 + WPCA + Random Forest achieved 88.01% accuracy, which is used as our reference benchmark.

Our Work: End-to-End CNN

- We remove the separate feature extraction and classical ML stages.
- A CNN is fine-tuned directly for benign vs. malignant classification.
- We compare ResNet50, DenseNet121, MobileNetV2, and EfficientNet-B4 under the same evaluation protocol.

Explainability

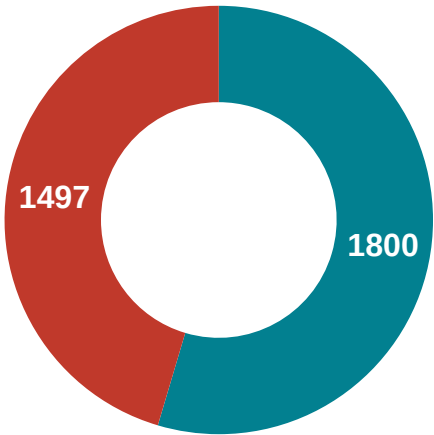
- We use **Grad-CAM** to visualize which parts of the lesion influence the model's prediction.
- This helps us understand the model's attention, but it is **not a clinical or causal explanation**.

Skin Cancer: Malignant vs. Benign (ISIC / Kaggle)

Stratified train / validation / test partitioning (seed = 42)

Subset	Benign	Malignant	Total
Training	1,152	957	2,109
Validation	288	240	528
Official Test	360	300	660
Total	1,800	1,497	3,297

Overall Class Balance (n = 3,297)



Benign Malignant

- The official test set is held out entirely; only the training split is further divided 80:20 into train / validation.
- Source: ISIC archive images distributed via Kaggle (“Skin Cancer: Malignant vs. Benign”).

Preprocessing & Augmentation

1

Decode & Resize

Images decoded as RGB and resized to 300×300 px.

2

Model-Specific Preprocessing

Each image is passed through the native preprocessing function of the selected CNN backbone.

3

Train-Only Augmentation

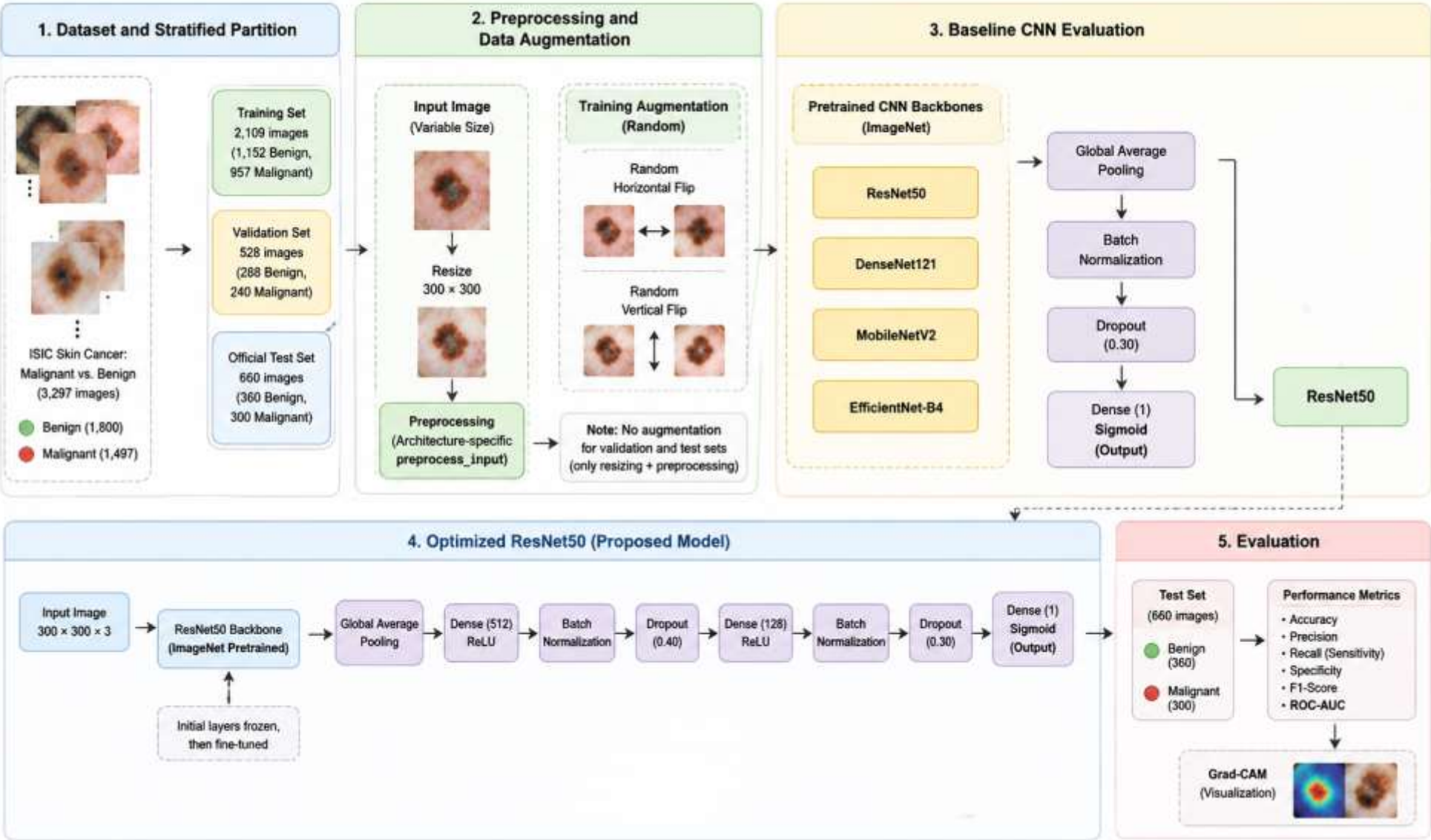
Random horizontal and vertical flips applied only to the training split.

4

Efficient Loading

tf.data pipeline with training-only shuffling, parallel mapping, batch size 32, and AUTOTUNE prefetch.

End-to-End Framework Overview



Comparing Four Pretrained CNN Architectures

BEST

ResNet50

Residual connections ease optimization of deep networks.

88.48%

Baseline Test Accuracy

DenseNet121

Dense connectivity promotes feature reuse across layers.

81.21%

Baseline Test Accuracy

MobileNetV2

Lightweight inverted residuals & linear bottlenecks.

84.85%

Baseline Test Accuracy

EfficientNet-B4

Compound scaling across depth, width & resolution.

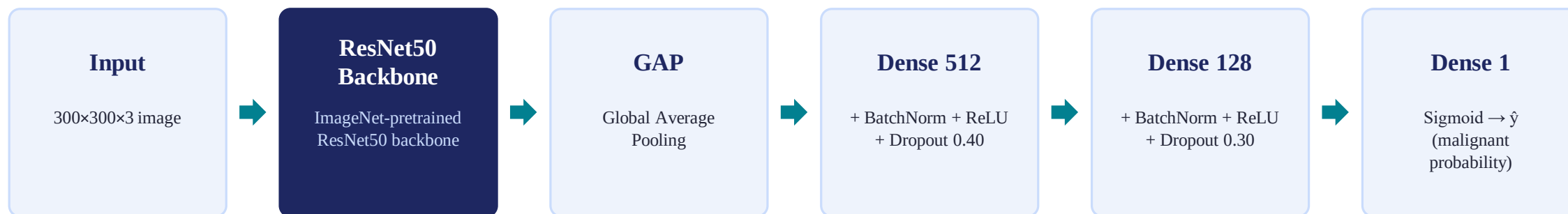
81.21%

Baseline Test Accuracy

Output Snapshot:

=====								
BASELINES VALIDATION PERFORMANCE COMPARISON (300x300 RESOLUTION)								
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	Model	Val Accuracy	Val Precision	Val Sensitivity	Val Specificity	Val Recall	Val F1-Score	Val ROC-AUC
	Baseline ResNet50	88.64%	87.19%	87.92%	89.24%	87.92%	87.55%	96.01%
	MobileNetV2 Baseline	85.04%	83.68%	83.33%	86.46%	83.33%	83.51%	92.83%
	DenseNet121 Baseline	77.46%	83.43%	62.92%	89.58%	62.92%	71.73%	88.67%
	EfficientNet-B4 Baseline	75.57%	82.84%	58.33%	89.93%	58.33%	68.46%	89.82%
=====								
Selected Strongest Baseline Architecture : ResNet50								

Fine-Tuned ResNet50 — Classification Head



KEY DESIGN CHOICES

Global Average Pooling

Reduces feature dimensions and makes the feature information simpler.

Dropout Regularization

Dropout rates of **0.40** and **0.30** help reduce overfitting.

Sigmoid Output

Produces the probability of the **malignant** class.

- Backbone features are globally pooled and passed through two regularized dense blocks (512 → 128) before the final sigmoid output.

Two-Stage Progressive Fine-Tuning

Stage 1 — Head Warm-Up

- ResNet50 backbone fully frozen
- Only the new classification head is trained
- Optimizer: Adam, learning rate = 1×10^{-3}
- Up to 15 epochs, EarlyStopping (patience = 4)
- Best validation weights restored



Stage 2 — Partial Fine-Tuning

- Top 30 backbone layers unfrozen
- The unfrozen layers and classification head are trained together.
- Optimizer: Adam, learning rate = 2×10^{-5}
- Up to 15 epochs, EarlyStopping (patience = 6)

Validation-Based Threshold Selection

- A fixed 0.50 threshold may not be optimal after transfer learning.
- Thresholds from 0.30 to 0.695 are tested in steps of 0.005 using validation data only.
- For each threshold, Accuracy, Precision, Sensitivity, Specificity, and F1-Score are calculated.
- The selected threshold is fixed before test evaluation; no test-set tuning is performed.

Composite Objective

$$S(T) = 5 \cdot m(T) + 2 \cdot A(T) + F1(T)$$

$$m(T) = \min\{A(T), P(T), Se(T), Sp(T), F1(T)\}$$

$$T^* = \text{argmax over } T \text{ of } S(T)$$

Gives more importance to the weakest metric while also considering Accuracy and F1-Score.

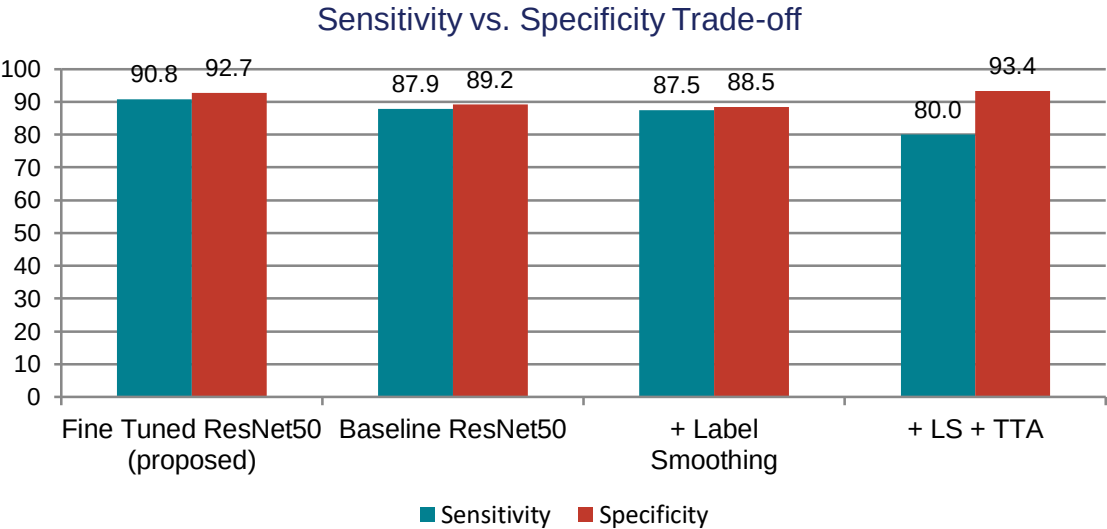
$$T^* = 0.495$$

Selected threshold for the final Fine-tuned ResNet50.
(Operating point ≈ 0.50)

Loss Function & Test-Time Augmentation

TABLE 2: ABLATION STUDY OF OPTIMIZED RESNET50 FRAMEWORK (VALIDATION SET)

Effect of Optimization Strategy, LS-BCE and Test-Time Augmentation									
Rank	Configuration	Threshold	Accuracy	Precision	Sensitivity	Specificity	Recall	F1-Score	ROC-AUC
1	2. Fine-tuned ResNet50	0.50	91.86%	91.21%	90.83%	92.71%	90.83%	91.02%	96.68%
2	1. Baseline ResNet50	0.50	88.64%	87.19%	87.92%	89.24%	87.92%	87.55%	96.01%
3	3. Fine-tuned ResNet50 with Label Smoothing	0.47	88.07%	86.42%	87.50%	88.54%	87.50%	86.96%	95.55%
4	4. Fine-tuned ResNet50 with Label Smoothing and Test-Time Augmentation	0.47	87.31%	91.00%	80.00%	93.40%	80.00%	85.14%	95.31%



- Label smoothing slightly reduces accuracy to 88.07% compared with the fine-tuned ResNet50 model.
- 5-view TTA increases specificity to 93.40%, but sensitivity drops to 80.00%, causing more malignant cases to be missed.
- Conclusion: The fine-tuned ResNet50 with standard BCE provides the best overall balance and is retained as the final configuration.

Grad-CAM: Visualizing Model Attention

- Grad-CAM is applied to the final convolutional layer, conv5_block3_out.
- Gradients from the malignant-class output are used to calculate the importance of each feature map.
- The weighted feature maps produce a heatmap showing important image regions.
- It is used for visual interpretation only, not as clinical proof.

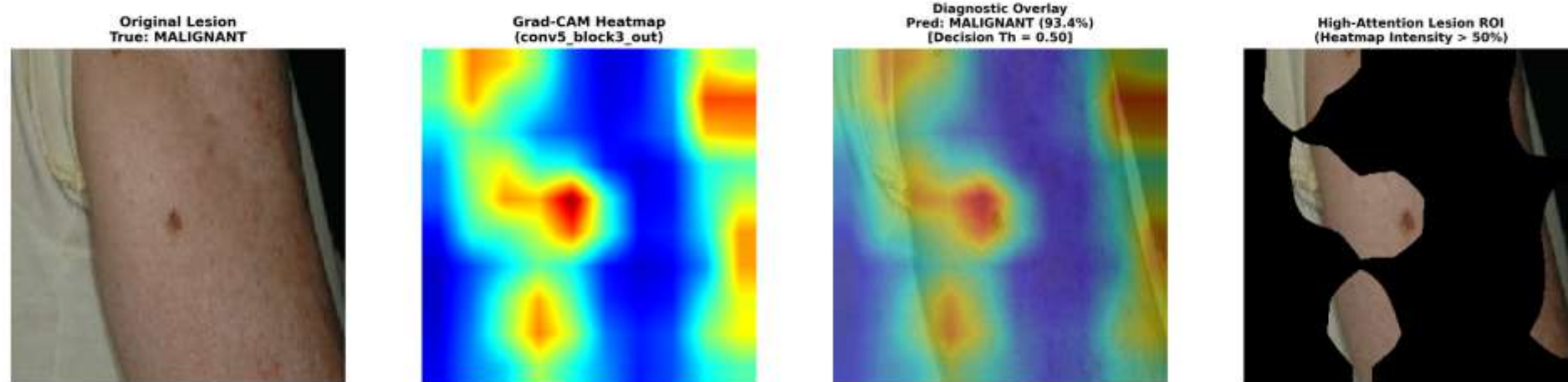
$$\alpha^{k^c} = (1/Z) \sum_i \sum_j \partial y^c / \partial A_{ij}^k$$

$$L^c = ReLU(\sum_k \alpha^{k^c} A^k) \quad \text{— Eq. (9)–(10)}$$

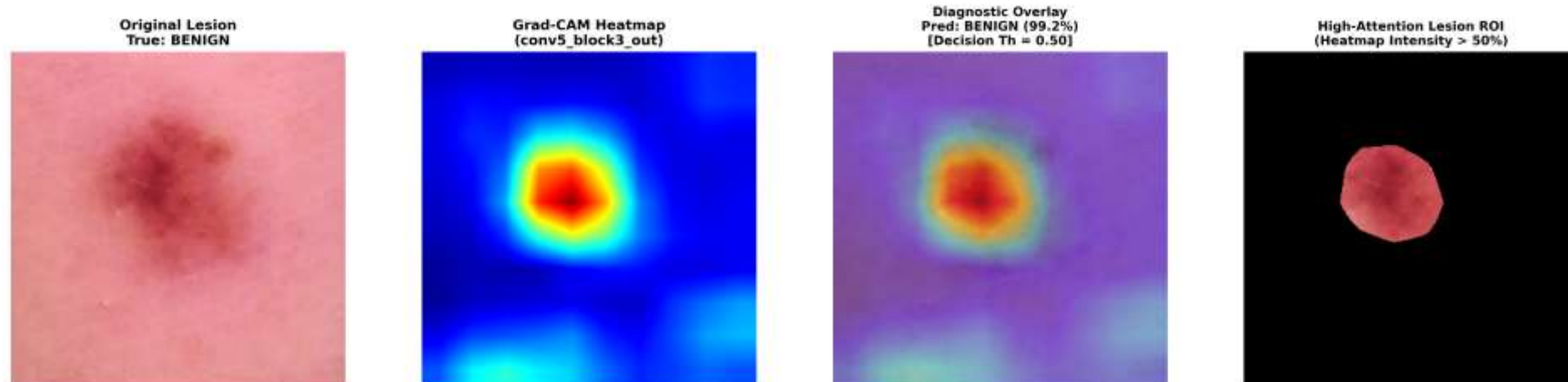
Channel weights α^{k^c} capture importance of each feature map for the malignant class y^c .

Grad-CAM: Visualizing Model Attention

Publication Explainable AI Panel — True Positive (Malignant Lesion)



Publication Explainable AI Panel — True Negative (Benign Lesion)



Official Benchmark Test-Set Performance

TABLE 1: BASELINE VS PROPOSED SINGLE MODEL COMPARISON
OFFICIAL BENCHMARK TEST SET

Performance Comparison of Baseline and Optimized Single Architectures									
Rank	Model	Threshold	Accuracy	Precision	Sensitivity	Specificity	Recall	F1-Score	ROC-AUC
1	Fine-tuned ResNet50	0.50	90.91%	88.22%	92.33%	89.72%	92.33%	90.23%	96.59%
2	Baseline ResNet50	0.51	88.48%	84.78%	91.00%	86.39%	91.00%	87.78%	95.40%
3	Fine-tuned ResNet50 with Label Smoothing	0.51	88.03%	85.76%	88.33%	87.78%	88.33%	87.03%	95.55%
4	Fine-tuned ResNet50 with Label Smoothing and Test-Time Augmentation	0.51	86.52%	91.05%	78.00%	93.61%	78.00%	84.02%	95.38%
5	MobileNetV2 Baseline	0.50	84.85%	84.01%	82.33%	86.94%	82.33%	83.16%	91.98%
6	EfficientNet-B4 Baseline	0.38	81.21%	73.78%	91.00%	73.06%	91.00%	81.49%	89.27%
7	DenseNet121 Baseline	0.37	81.21%	76.19%	85.33%	77.78%	85.33%	80.50%	89.14%

FINAL PROPOSED MODEL SUMMARY

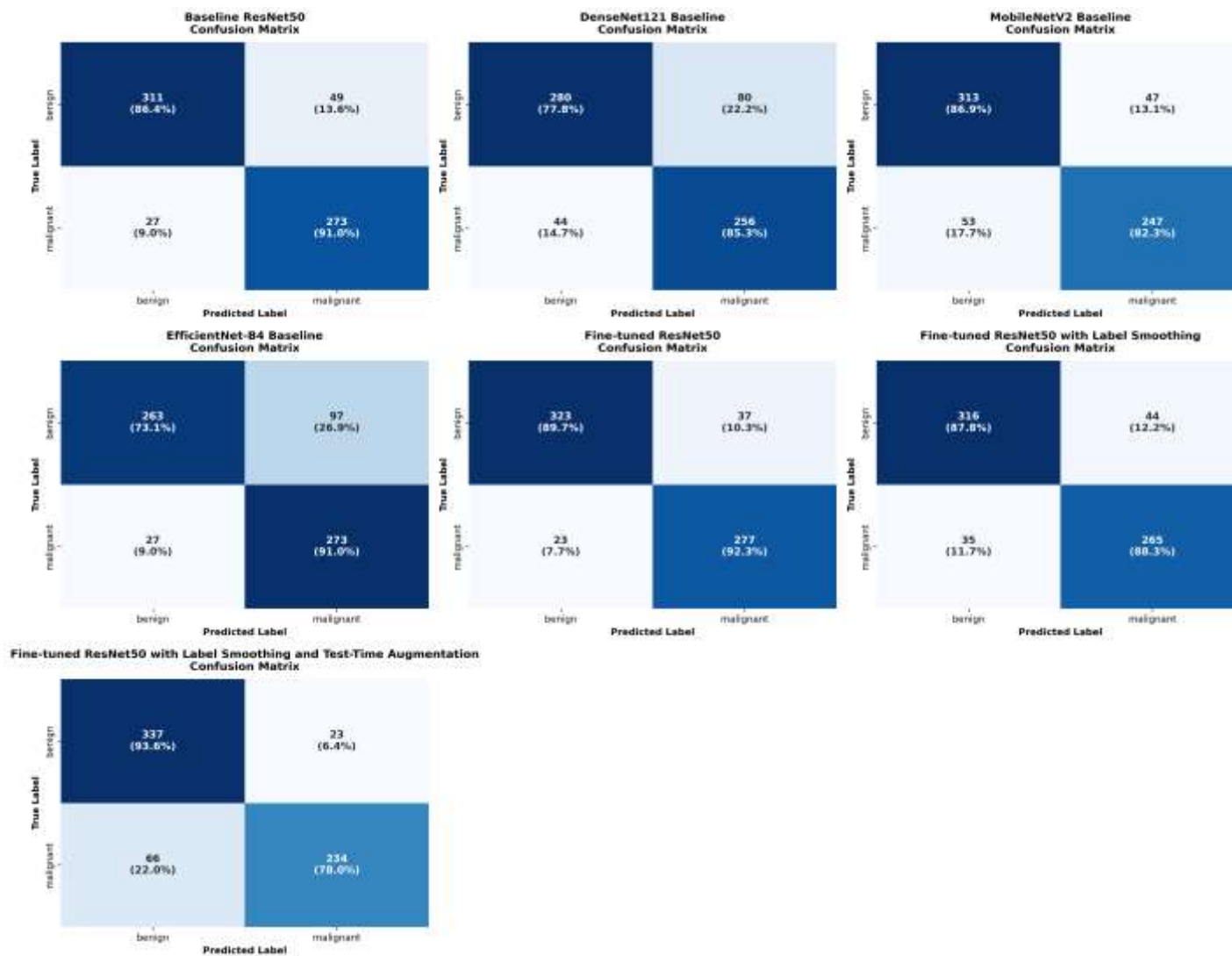
Final Proposed Model Performance							
Selected Model	Threshold	Accuracy	Precision	Sensitivity	Specificity	F1-Score	ROC-AUC
Fine-tuned ResNet50	0.495000	90.91%	88.22%	92.33%	89.72%	90.23%	96.59%

Official Benchmark Test-Set Performance

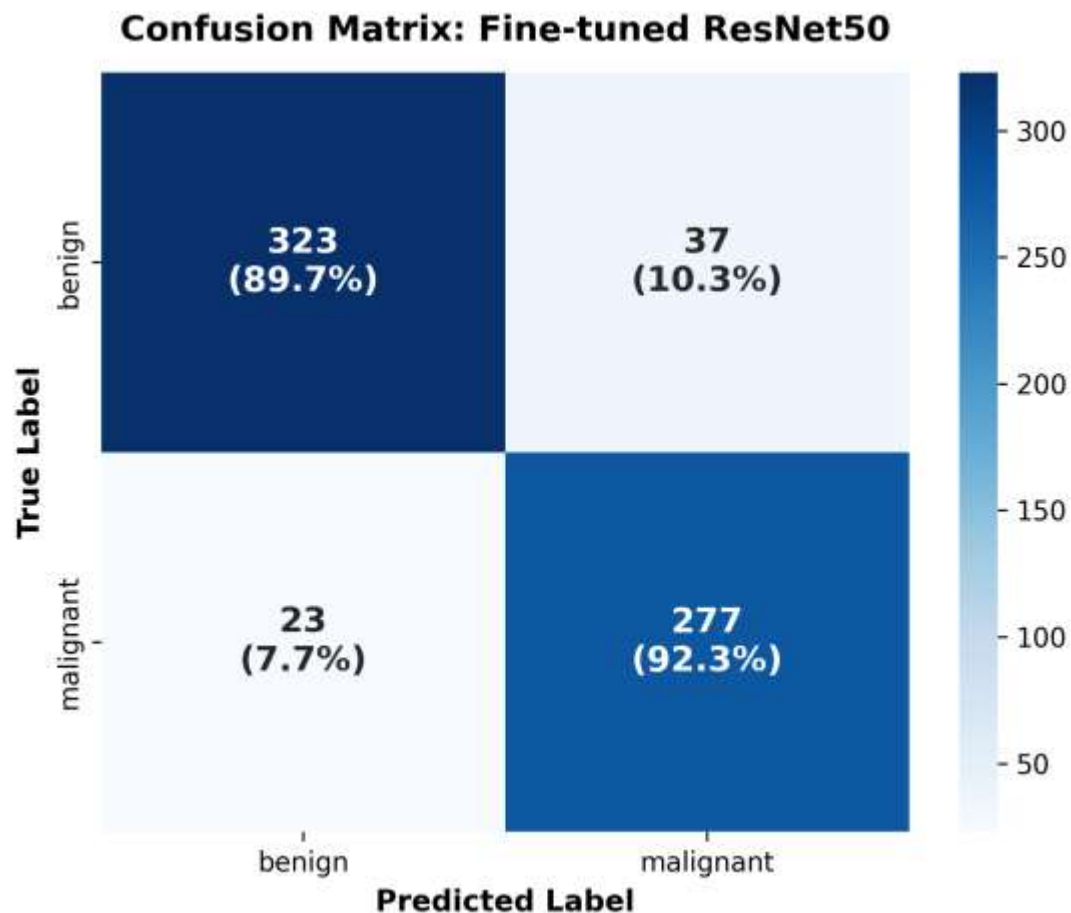
Model	T*	Acc.	Prec.	Sens.	Spec.	F1	AUC
Fine-tuned ResNet50	0.50	90.91%	88.22%	92.33%	89.72%	90.23%	96.59%
Baseline ResNet50	0.51	88.48%	84.78%	91.00%	86.39%	87.78%	95.40%
FT ResNet50 + LS	0.51	88.03%	85.76%	88.33%	87.78%	87.03%	95.55%
FT ResNet50 + LS + TTA	0.51	86.52%	91.05%	78.00%	93.61%	84.02%	95.38%
MobileNetV2 Baseline	0.50	84.85%	84.01%	82.33%	86.94%	83.16%	91.98%
EfficientNet-B4 Baseline	0.38	81.21%	73.78%	91.00%	73.06%	81.49%	89.27%
DenseNet121 Baseline	0.37	81.21%	76.19%	85.33%	77.78%	80.50%	89.14%

Confusion Matrix — All Models

Benchmark Test Set Confusion Matrices Across All Evaluated Models



Confusion Matrix — Fine-Tuned ResNet50 (Best Model)



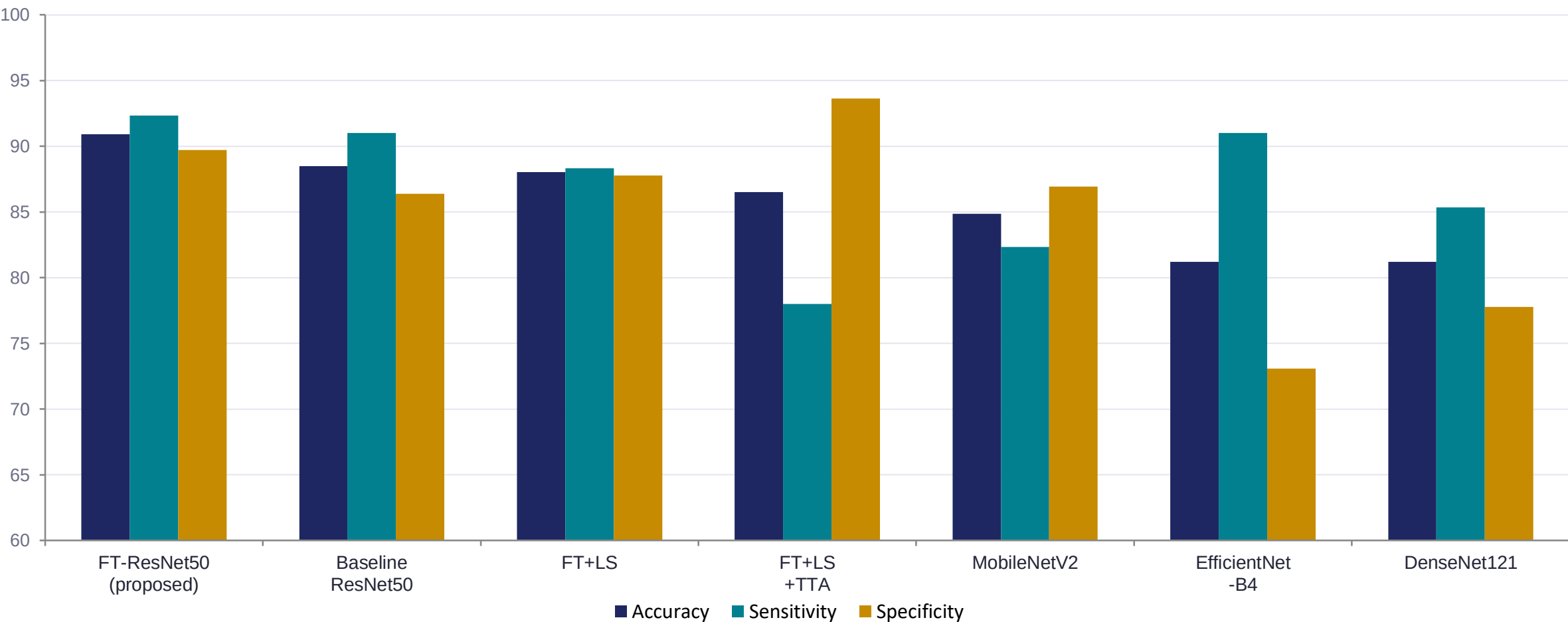
92.33% Sensitivity (Recall)

89.72% Specificity

88.22% Precision

96.59% ROC-AUC

Performance Across All Evaluated Models



Comparison with Published State-of-the-Art

Author & Ref.	Year	Model / Method	Acc. (%)	Sens. (%)
Farooq et al.	2020	InceptionV3	86.00	89.00
Ismail et al.	2021	ResNet	75.31	–
Kartal & Polat	2022	ResNet152	88.79	87.71
Hussein et al.	2023	SqueezeNet	85.90	80.33
Bazgir et al.	2024	Inception Network	85.94	86.89
Belabbaci et al.	2025	DarkNet53 + WPCA + RF	88.01	87.69
Proposed	2026	Fine-tuned ResNet50	90.91	92.33

Reference Benchmark

DarkNet53 + WPCA + RF

88.01% Accuracy · 87.69% Sensitivity

Proposed

Fine-tuned ResNet50

90.91% Accuracy · 92.33% Sensitivity

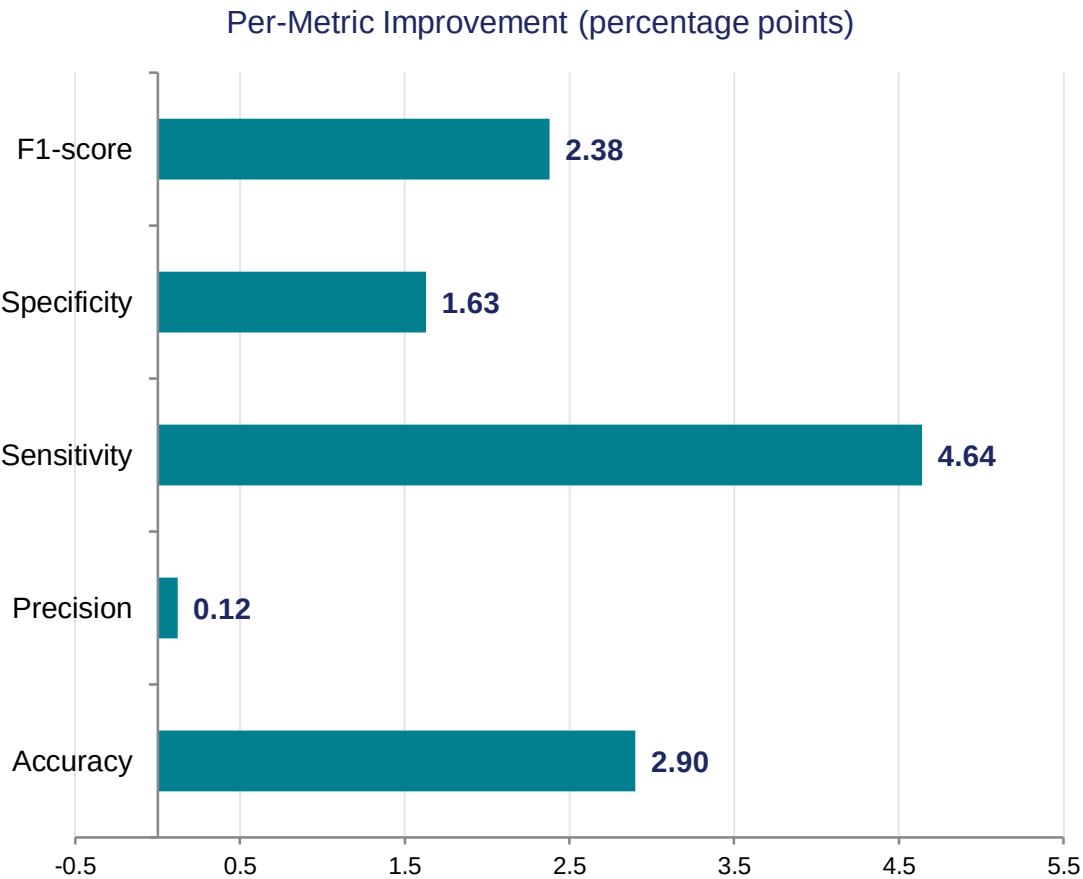
+2.90 pp

Accuracy improvement over strongest published baseline

Gain Over the Reference Framework

Metric	Reference (%)	Proposed (%)	Gain (pp)
Accuracy	88.01	90.91	+2.90
Precision	88.10	88.22	+0.12
Sensitivity	87.69	92.33	+4.64
Specificity	88.09	89.72	+1.63
F1-score	87.85	90.23	+2.38

ROC-AUC is omitted from this comparison — the reference study does not report an AUC value for its final DarkNet53 + WPCA + RF configuration.



Key Contributions & Takeaways

- Compared four pretrained CNN models (ResNet50, DenseNet121, MobileNetV2, EfficientNet-B4) under the same 300×300 input setting.
- Optimized ResNet50 with a two-layer dense head, staged fine-tuning, and validation-only threshold calibration.
- Direct ablation of BCE vs. label-smoothed BCE vs. 5-view TTA — showing extra regularization is not automatically beneficial.
- Full evaluation (accuracy, precision, sensitivity, specificity, F1, ROC-AUC, confusion matrices) plus qualitative Grad-CAM analysis.

FINAL RESULT

Fine-tuned ResNet50

90.91% Accuracy

92.33% Sensitivity

96.59% ROC-AUC

+2.90 pp Accuracy gain over the best published benchmark

Thank You

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